PROBLEM STATEMENT #29 | Smart Cities, Transport & Logistics

Municipal Infrastructure Damage Reporting App

1. Problem Context & Overview

A citizen portal for reporting road potholes, broken streetlights, and water pipeline leaks using geotagged photos, with automated ticket dispatch to relevant municipal engineering departments.

Target Stakeholders / Actors: *Citizen Reporter, Municipal Engineer*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]
Description: The system shall extract EXIF GPS metadata from uploaded damage photos to automatically pinpoint incident location on the municipal GIS map.
Sample Acceptance Criteria: Pass: Pothole report routed to Road Infrastructure division with exact GPS coordinates; Fail: Report submitted without location data.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]
Description: The reporting portal shall support offline report drafting and auto-sync when mobile network connectivity resumes.
Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**
  - >> Exactly 5 Functional Requirements (FR-001 to FR-005)
  - >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with
    ID, Type, Description, Priority, Acceptance Criteria, and Rationale.
- **UML Use-Case Diagram:**
  - >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.
- **Use-Case Flow Specification:**
  - >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
  - >> One Alternate Flow.